Parand Khalili

parandkhalili@aut.ac.ir | LinkedIn | GitHub | Website

Education

Amirkabir University of Technology (Tehran Polytechnic) | Sep 2022 - Present

B.Sc. in Computer Engineering | GPA: 18.47/20

- Undergraduate thesis: A Structure-Aware LLM-Based Framework for Graph Node Labeling.

Research Interests

- **Graph Machine Learning:** Representation learning and node classification on text-attributed graphs; combining structural and semantic signals.
- **LLM-Augmented, Data-Efficient Learning:** Pseudo-label generation, active sample selection, and learning with limited human annotations.
- **Reliable ML and Evaluation:** Failure analysis, uncertainty in LLM-generated labels, calibration, and leakage-aware experimental design.
- **Data Mining and Applied ML:** Pattern discovery and predictive modeling for real-world, decision-oriented datasets.

Research Experience

Undergraduate Thesis Research - LLMs for Graph Node Classification | 2026 - Present

Supervisor: Dr. Mostafa Haghir Chehreghani, Amirkabir University of Technology

- Implemented and extended an LLM-to-GNN node-classification workflow inspired by published label-free methods; examined prompts incorporating real one- and two-hop graph evidence on Cora and CiteSeer.
- Conducted controlled prompt comparisons, examining pseudo-label accuracy, confidence, and downstream GCN behavior; currently preparing a matched FeatProp-based evaluation.

Related Investigation - LLM Annotation Failures and Reliability | Ongoing

Exploratory direction emerging from the undergraduate thesis

- Examining failed and conflicting LLM node annotations, confidence limitations, and evaluation pitfalls; developing an interface to inspect thesis-workflow outputs (in progress).

Honors & Awards

National University Entrance Examination (Konkour), Iran | 2022

Ranked 134 among more than 150,000 participants nationwide.

Teaching Experience

Head Teaching Assistant - Business Intelligence | Fall 2025

Instructor: Dr. Ahmadi

- Developed teaching materials and course projects; taught Power BI, Python, SQL, and analytics data modeling.

Data Mining | Dr. Nazerfard | Spring 2026

Software Engineering II (Advanced) | Dr. Gohari | Spring 2026

Theory of Languages & Automata | Dr. Zakeri | Fall 2025

Software Engineering I | Dr. Alirezaei | Spring, Fall 2025

Applied Linear Algebra | Dr. Haghir Chehreghani / Dr. Nazerfard | Fall 2024; Spring/Fall 2025

Microprocessor & Assembly Language | Dr. Farbeh | Fall 2025

Computer Architecture | Dr. Farghadan | Spring 2025

Advanced Programming | Dr. Zeinali | Spring 2024

Professional Experience

Retail Data Analyst - EMRC | Jul 2026 - Present

- Develop analytical dashboards and client-facing reports using real-world retail data, translating business requirements into performance indicators and actionable insights.
- Collaborate directly with clients to clarify reporting needs, communicate findings, and refine analytical deliverables; contribute to the ongoing automation of recurring reporting workflows.
- Independently developed a retail sales forecasting system using statistical methods, CatBoost, and ensemble models, with feature engineering, historical backtesting, and WAPE-based evaluation across forecast horizons.
- Built an interactive forecasting interface for configuring predictions, visualizing results, and exporting reports; analyzed model errors across product and geographic segments.

Data Analysis & BI Intern - YAS Arghavani System Engineering Co. | Summer 2025

Bank Mellat IT Holding

- Built Power BI dashboards to monitor POS performance, maintenance, and operational efficiency.
- Modeled transactional data using Power Query and DAX for revenue KPIs; integrated CRM and payment-system data to improve reporting access.

Selected Technical Projects

Credit Scoring & Loan Default Risk Mitigation | 2026

Python, scikit-learn

- Built an ML pipeline with target encoding and engineered financial features to address class imbalance; tuned a Random Forest model achieving 87.41% accuracy, 79.21% balanced accuracy, and 4.65 decile lift in the reported evaluation.

Artificial Intelligence Portfolio | 2025

Python, NumPy

- Implemented search and game-playing agents (A*, minimax/alpha-beta), constraint solving (CSP/AC-3), Q-learning, and Bayesian-network inference (Gibbs sampling and likelihood weighting).

MyTorch - Neural Network Components | 2025

Python, neural networks

- Implemented tensor operations, backpropagation, activation functions, and stochastic gradient descent in an educational neural-network framework.

Tetris AI via Genetic Algorithms | 2025

Python, Pygame

- Developed an autonomous Tetris agent with evolutionary selection, crossover, and mutation; designed a board-evaluation fitness function and visualized training.

xv6 Operating System Enhancements | 2025

C, Linux, Docker

- Added custom system calls and multithreading support, working with memory management and process scheduling in xv6.

Technical Skills

Programming: Python, C++, C, Java, SQL

Machine Learning & Research: PyTorch, PyTorch Geometric, scikit-learn, GCNs, graph node classification, active learning, LLM prompting, experimental evaluation

Data & BI: NumPy, Pandas, Matplotlib, Power BI, DAX, Power Query, data modeling

Development Tools: Git, Linux, Docker

Certifications

Data Analysis with Python, Quera | Coursework: Summer 2025; certificate issued March 2026 (perfect score)

Advanced Python, Sematec | 2024 | 40 hours | Score: 100

Languages & Test Scores

Persian (native) | English: TOEFL iBT 5/6 (CEFR C1)
Reading 5, Listening 6, Writing 5, Speaking 4.5